Fiscal Year:	FY 2019	**Task Last Updated:**	FY 05/21/2019
PI Name:	Mason, Christopher Ph.D.		
Project Title:	An Integrated Framework for Characterization of the Noncoding Genome and Epigenome in Astronauts		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	chm2042@med.cornell.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	203-668-1448
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City:	New York	**State:**	NY
Zip Code:	10021-5663	**Congressional District:**	12
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2015-16 HERO NNJ15ZSA001N-Crew Health (FLAGSHIP, NSBRI, OMNIBUS). Appendix A-Crew Health, Appendix B-NSBRI, Appendix C-Omnibus
Start Date:	10/26/2016	**End Date:**	10/25/2018
No. of Post Docs:	3	**No. of PhD Degrees:**	2
No. of PhD Candidates:	3	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:	Norsk, Peter	**Contact Phone:**	
Contact Email:	Peter.norsk@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 10/25/2018 per NSSC information; original end date was 10/25/2017 (Ed., 1/16/18)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Snyder, Michael Ph.D. (Stanford University)		
Grant/Contract No.:	NNX17AB26G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	(1) Summary: Massively parallel DNA sequencing continues to provide astonishing insights into the regulatory complexity of the mammalian genome, revealing heritable and de novo occurring risk factors predisposing humans to a wide variety of diseases and disorders. Notably, members of the ENCyclopedia of DNA Elements (ENCODE) Consortium, including Co-Investigators Drs. Mason and Snyder, have recently mapped thousands of non-coding, regulatory elements in the human genome that specify risk and protective sites for many diseases. These noncoding, genetic variations also show differential impact depending on the epigenetic changes of the genome, including methylation of DNA and chromatin states inside cells. Yet, these questions have not been comprehensively examined in human astronauts, due to the fact that the very first data sets that can delineate the putative impact of genome-wide noncoding variation and epigenetic changes from space travel are only now being generated in the NASA Twins Study. However, with this data from NASA Twins study, there is a unique opportunity to both generate new computational tools for the integration of the data from these samples and also to utilize novel, single-cell approaches for epigenetic states in the human cells taken from space. We propose to expand and further integrate our tools for examining the regulatory changes to DNA and RNA, together called “EpiSuite,” which will provide long-sought solutions to problems in integrative genomics and establish a more systems-level view of human physiology during space travel. Specifically, we will optimize and integrate open-source and published bioinformatics tools between both laboratories (HUGE-Seq, methylKit, eDMR, r-make), expand our code base with two new tools (methClone and CemBase), and then create a web-interface for users to upload, organize, and analyze these data relative to ENCODE public data. Then, we will use novel single-cell experimental methods (scATAC-seq) to confirm and characterize the impact and scope of such noncoding and epigenetic changes. Together, the data and tools in EpiSuite, concomitant with the new single-cell chromatin data, will enable discovery and validation of specific loci that represent the genetic and epigenetic risks faced by astronauts and also an improved understanding of the regulatory and genetic impact (BMED5) of these molecular changes. (2) Hypothesis: We hypothesize that specific areas of the genome are epigenetically dysregulated as a result of space travel, and that some of these areas differ between astronauts. Moreover, we hypothesize that the enrichment of non-coding variants in distinct biological pathways will also differ between astronauts, and analysis and integration of these data will reveal the first regions of “epigenetic space risk” for the astronauts, delineated in BMED5. To validate these predictions, we will perform single-cell ATAC-seq (Assay for Transposase Accessible Chromatin) on cell fractions from Earth-based control samples and those taken from space. (3) Specific Aims Aim 1. Create a web-based framework for integrating non-coding genomic variation and differential epigenetic states. This EpiSuite interface and set of machine-learning algorithms will help ongoing work with the NASA Twins study and aid future investigations that leverage multi-omics data sets from astronauts. These data will also be linked to gene expression data to gauge the impact of genetic and epigenetic changes on specific biological networks. Aim 2. Validate the altered chromatin states in cells from space travel and measure the proportion of cells changed with single-cell ATAC-seq. We will use an assay called single cell (scATAC-seq), which can measure the sites of open and closed chromatin on a cell-by-cell basis. This will validate the sites where we observe epigenetic changes and also provide the first estimate of the proportion of cells epigenetically impacted by space travel.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	As proposed in our grant and noted in Task Description above, we performed continued characterization of the NASA Twins Samples as well as published new methods, analyses, and techniques in integrative genomics and epigenetics, while engaging the public and broader scientific community around these efforts. Significant accomplishments: • Developed and published the mCaller epigenetics algorithm online; the paper is now published at Nature Communications (see Bibliography section--McIntyre et al., 2019) • Created a Spaceflight-Omics Browser for longitudinal expression and epigenetic data, used by the other Twins PIs and NASA researchers on the NASA Twins Study • Adjsuted APOBEC enzyme ratios for de-amination protocol for WG-nonBS, which is now called EM-seq • All backup DNA/RNA transferred to Stanford for sample backup • Completed age analysis of the whole-genome bisulfite sequencing (WGBS) data and developed a new “epigenetic age panel” for testing on the astronauts • Collected >18,000 data sets for reduced representation bisulfite sequencing (RRBS), 450K, 850K, and WGBS data for controls • Completed STRINGTIE analysis for ncRNA, circRNA, and cfRNA mapping • Validated newly predicted circRNAs and predict the cell type of origin, based on gene expression signatures • Analyzed the miRNA for potentially novel “Space miRNAs.”
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